MINI-PROJECT · B.TECH CSE (AI & ML) · MANIPAL INSTITUTE OF TECHNOLOGY

Machine-Learning Trading Signals

A comparative study against technical baseline strategies

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School of Computer Engineering · 2026



SPY · daily adjusted close · 2010–2025 (data used in this study)

THE BIG IDEA

Can a model learn when to be in the market?

RULE-BASED

```
if SMA(10) > SMA(50):   long
else:                   cash
```

Written by hand and fixed forever: the classic moving-average crossover baseline.

vs

LEARNED

```
if P(Up | 42 features) ≥ τ:   long
else:                         cash
```

Five model families learn P(Up) from 2011–2019 data; the threshold τ is chosen on 2020–2021.

Same data · same execution · same costs — so any difference comes from **the signal itself.**

Tested on SPY (an S&P 500 ETF), Apple (AAPL) and Microsoft (MSFT): daily prices 2010–2025 from Yahoo Finance.



Four easy ways to fool yourself in a backtest

01

Look-ahead bias

Features or labels that quietly use tomorrow's data, so the backtest trades on information nobody had.

→ too-good accuracy

02

Leaky validation

Random K-fold on a time series lets the model train on the future of the days it is tested on.

→ optimistic CV scores

03

Free trading

Ignoring fees, slippage and when orders actually fill. High-turnover strategies die on costs.

→ paper-only profits

04

Weak baselines

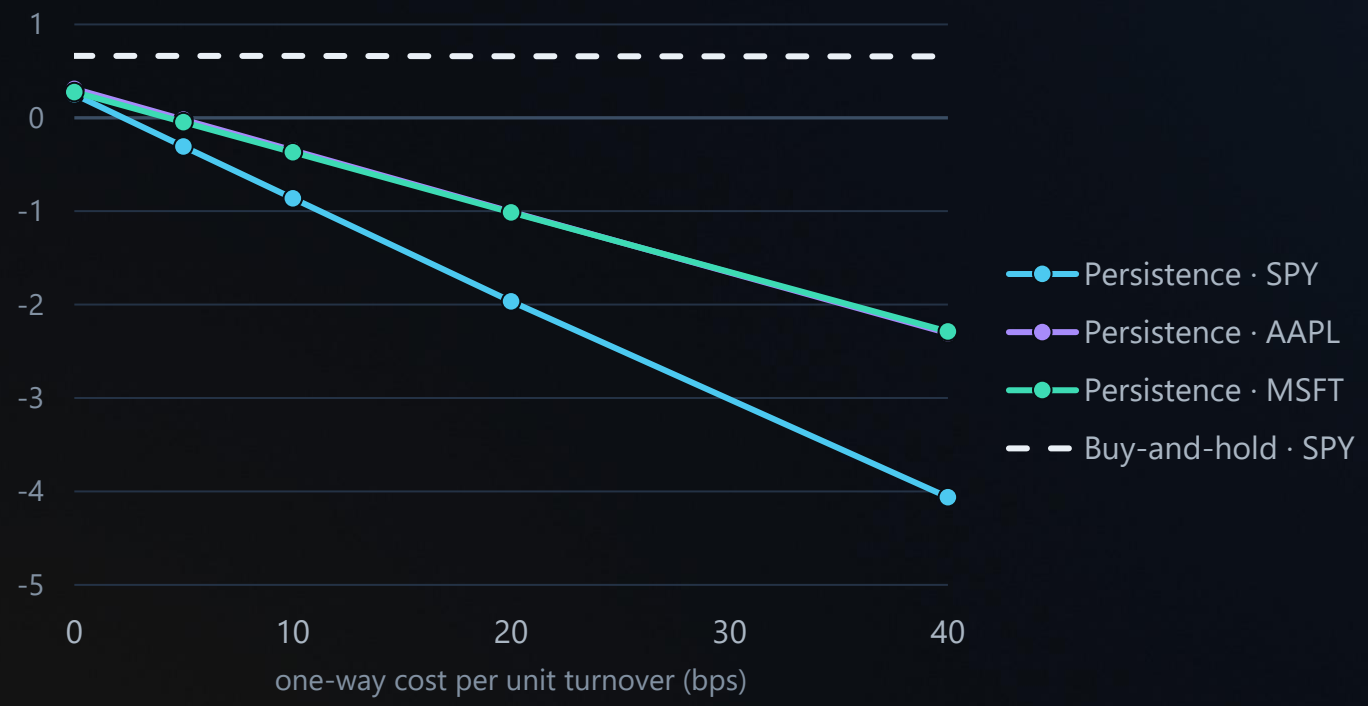
A model at 51% sounds better than a coin flip, but 'always predict Up' already scores 49.5% on SPY.

→ base-rate 'skill'

Each trap inflates results. The framework was designed to close all four.

Small evaluation choices flip the conclusion

Naive persistence — Sharpe ratio vs. one-way cost (bps), test 2022–2025



Persistence ('tomorrow repeats today') is profitable before costs and deeply negative after them. AAPL and MSFT nearly coincide.

0.58 → 0.01

AAPL Random Forest Sharpe when fills move from the next open to the same day's close (sensitivity run)

−0.21 … 1.58

95% bootstrap interval for SPY buy-and-hold's Sharpe ratio (0.66) over the four test years

Sharpe ratio = annualised mean return ÷ annualised volatility (risk-free rate 0). 1 bp (basis point) = 0.01%.

What we set out to do

01	Build moving-average crossover and buy-and-hold baselines in one backtester with realistic transaction costs.	DELIVERED
02	Engineer SMA, EMA, RSI, MACD and Bollinger-band features for XGBoost and LSTM directional signals.	DELIVERED
03	Compare ML signals with the baselines on predictive and risk-adjusted metrics across market conditions.	DELIVERED
04	Stretch goal: test whether news sentiment found by retrieval-augmented generation (RAG) makes the signals more robust.	FUTURE WORK

BEYOND THE SYNOPSIS Logistic Regression · Random Forest · soft-vote ensemble · walk-forward evaluation · bootstrap CIs · live dashboard

A leakage-safe, cost-aware comparison framework

01 Look-ahead bias	**Point-in-time features.** Scale-invariant ratios only; tests recompute them on truncated data and require identical values.
02 Leaky validation	**Chronological splits.** Train → validation → test with a 2-day purge; purged CV for the tabular models, a purged hold-out slice for the LSTM.
03 Free trading	**Realistic execution.** Decide at the close, fill at the next open, and pay 15 bps (10 fee + 5 slippage) on every unit of turnover.
04 Weak baselines	**Baselines and intervals.** Majority class, persistence, MA crossover and buy-and-hold, with block-bootstrap 95% CIs.

Every main result uses all four safeguards; sensitivity runs relax one at a time and are labelled.

System architecture: from raw prices to decisions

01 Data
Yahoo Finance OHLCV, validated and adjusted
`data_loader.py`

→

02 Features
42 point-in-time, scale-invariant indicators
`features.py`

→

03 Targets
Down / Neutral / Up from next-open returns
`targets.py`

→

04 Split
Chronological train / val / test + purge
`preprocessing.py`

↓

05 Models
LR · RF · XGBoost · LSTM · ensemble
`models.py · lstm_model.py`

→

06 Signal rule
τ chosen on validation; walk-forward runner
`experiment.py`

→

07 Backtester
Next-open fills, 15 bps per unit turnover
`backtester.py`

→

08 Evaluation
Metrics, bootstrap CIs, calibration
`evaluation.py`

↓

OUTPUTS result tables · 46 figures · experiment report · IEEE report · interactive dashboard `reporting.py · report/ · app/`

Built on the open-source Python stack

DATA

Acquire & engineer

- Python
  3.14.3
- pandas
  2.3.3
- NumPy
  2.4.3
- yfinance
  Yahoo Finance OHLCV

MODELLING

Learn P(Up)

- scikit-learn
  1.8.0
- XGBoost
  3.2.0
- PyTorch
  2.11.0 · CPU

EVALUATION

Measure honestly

- SciPy
  statistics
- matplotlib
  46 figures
- pytest
  301 tests

DELIVERY

Show & share

- Streamlit
  interactive dashboard
- Plotly
  dashboard charts
- Jupyter
  3 executed notebooks
- python-docx
  IEEE report builder

Versions as recorded for the reported run (results/metrics/run_summary.json). One configuration file, config.yaml, with a fixed seed (42).

Time only flows forward: chronological splits



TRAIN · 2011–2019

Fit every model. Tabular models are tuned with purged time-series CV (4 folds, gap 2) on log loss; the LSTM on a purged hold-out slice.

VALIDATION · 2020–2021

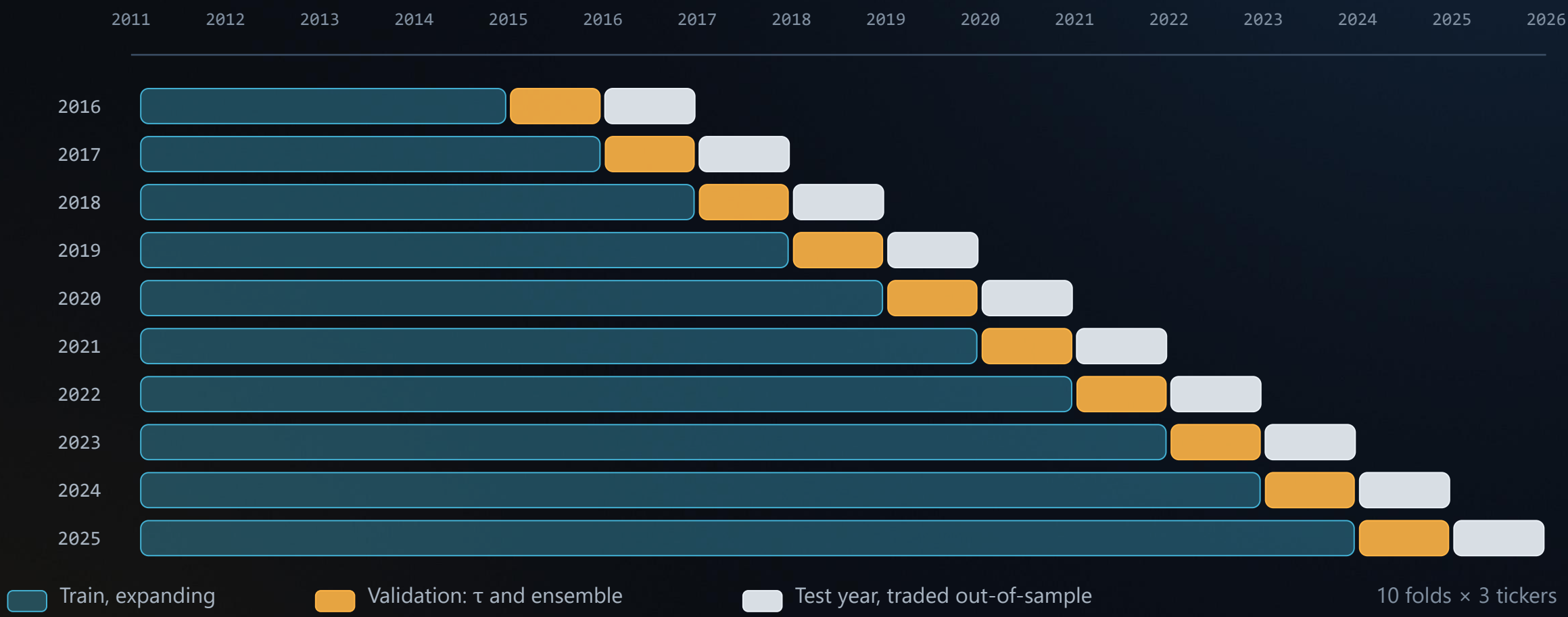
Choose ensemble members (log loss) and the threshold τ from 14 values (Sharpe).

TEST · 2022–2025

Never used for fitting or for choosing τ. Main results come from here; robustness checks re-use it and are labelled.

Rows are never shuffled. Each label looks two opens ahead, so the last two rows before every boundary are purged from the earlier segment.

Walk-forward: retrain every year, trade the next



42 point-in-time features, one executable target

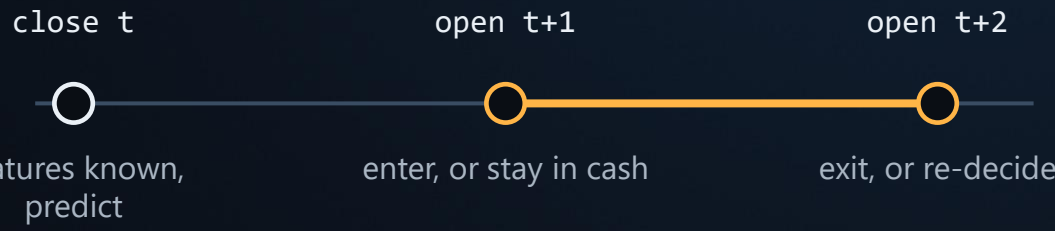
42 FEATURES · 5 GROUPS



- **Returns · 14** — lags 1, 2, 3 and 5 days, momentum 5–60 days, overnight and intraday moves, range
- **Technical · 15** — close/SMA and close/EMA (5–50), RSI(14), MACD, Bollinger %B and width
- **Volatility · 6** — ATR(14), realised volatility 5/20/60 days, vol ratio, normalised return
- **Volume · 3** — relative volume (5 and 20 days), volume z-score
- **Regime · 4** — volatility percentile, close/SMA200, golden cross, 252-day drawdown

Only ratios, never price levels, so back-adjusted prices cannot leak the future. A test recomputes features on truncated data and requires identical values.

TARGET · WHAT THE MODEL PREDICTS



$$r_t = O_{t+2} / O_{t+1} - 1$$

DOWN	NEUTRAL	UP
$r < -0.1\%$	$\lvert r \rvert \le 0.1\%$	$r > +0.1\%$

SPY test days: Up 49.5% · Down 41.2% · Neutral 9.4%

Five model families, tuned without peeking

LR

Logistic Regression

Linear, L2-regularised

SELECTED

```
C = 0.001
all three tickers
```

RF

Random Forest

300 bagged trees

SELECTED

```
depth 3
min leaf 80/20/80
```

XGB

XGBoost

Gradient-boosted trees

SELECTED

```
depth 2
η 0.03
100 trees
```

LSTM

LSTM network

PyTorch, 1 layer

SELECTED

```
hidden 32/16/16
window 10 days
```

ENS

Soft-vote ensemble

Two members, chosen on validation

SELECTED

```
SPY: LR + LSTM
AAPL: RF + XGB
MSFT: LR + RF
```

Where values differ they are shown as SPY / AAPL / MSFT. Tabular models are tuned on log loss with purged time-series CV; the LSTM is tuned and early-stopped on a purged slice at the end of the training period.

INSIGHT Tuning picked the most regularised option in almost every grid: there is little stable signal to fit.

BASELINES majority class · naive persistence · MA crossover (10/50) · buy-and-hold

From probability to position, net of costs

DECISION RULE



- Long at the next open when P(Up) ≥ τ, otherwise cash (long-only).
- τ comes from 14 values (0.30–0.65) and is chosen by validation Sharpe, never on test.
- Chosen τ ranged 0.30–0.45 across the 15 model–ticker pairs.
- Also backtested: an argmax rule (long when Up is the most likely class).

BACKTEST · EACH DAY t

$$cost_t = c \cdot |p_t - p_{t-1}| \cdot E_{t-1}$$
$$E_t = (E_{t-1} - cost_t) \cdot (1 + p_t\, r_t)$$

p = position (0 or 1) · r = next-open return · E = equity · c = 15 bps per unit turnover (10 fee + 5 slippage)

REPORTED FOR EVERY STRATEGY

cumulative return · annual return · volatility · Sharpe · Sortino · max drawdown · Calmar · win rate · turnover · trades · costs paid

Engineered like research software

301

automated tests

pytest, including look-ahead and truncation-invariance checks

1

command for all results

`python -m scripts.run_experiment`

24 min

full experiment on a CPU

seeded and deterministic (seed 42)

46

figures, generated

plus a Markdown experiment report; the IEEE report has its own builder

6

dashboard tabs

Streamlit app with live re-training on fresh data

2,000

bootstrap resamples per CI

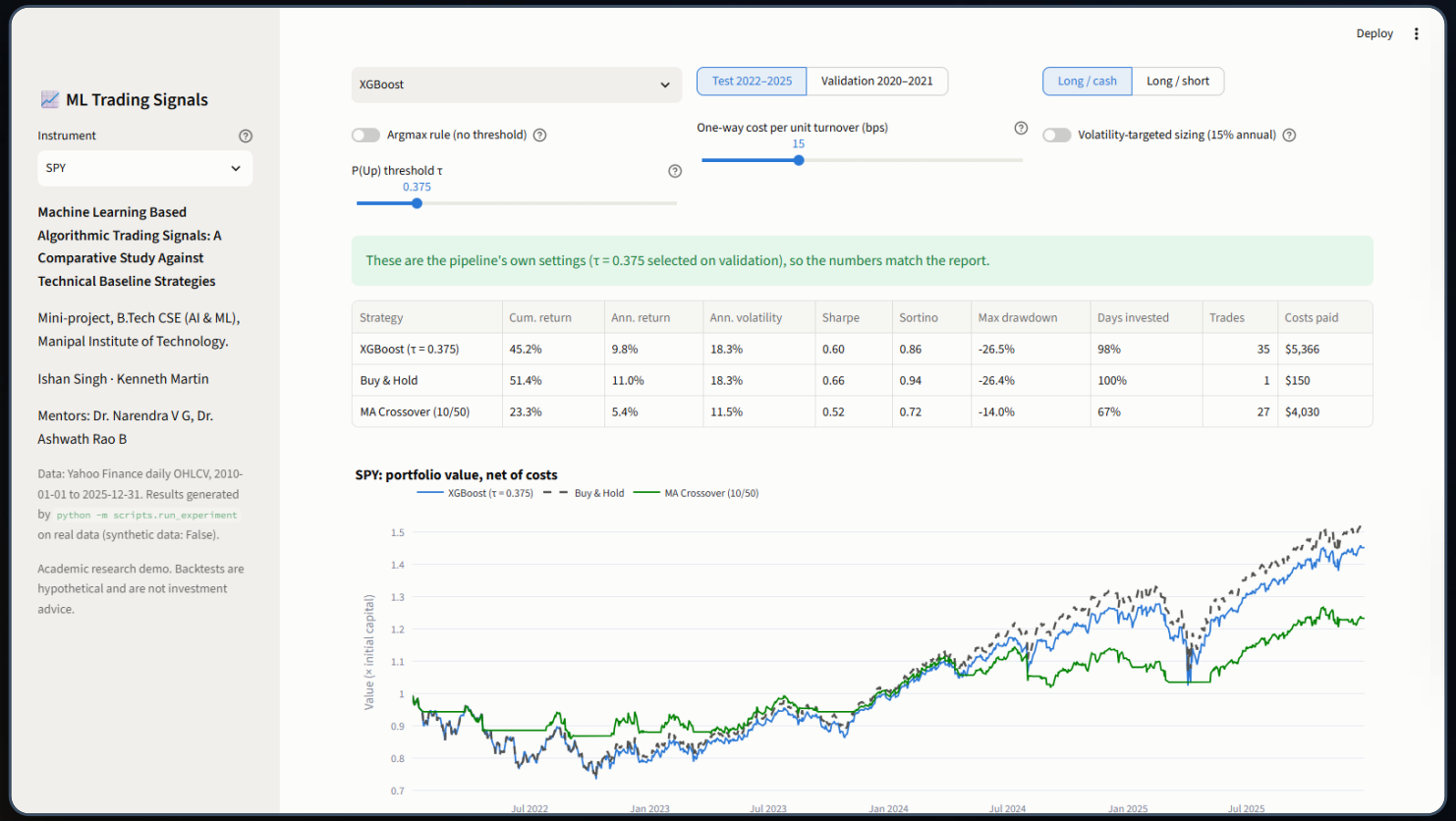
circular block bootstrap, block length 20

All counts are measured from the repository: tests collected by pytest, figures in results/figures, runtime from results/metrics/run_summary.json.

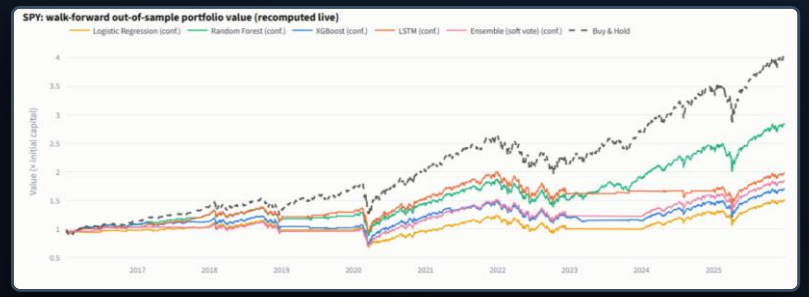
REAL SCREENSHOTS · LOCALHOST:8501 · CAPTURED 24 OCT 2026

The working system: an interactive dashboard

INTERACTIVE BACKTEST model · τ · costs · sizing, recomputed live



WALK-FORWARD 2016–2025



Dashed: buy-and-hold, finishing above all five ML strategies (SPY, walk-forward).

TRAIN A MODEL LIVE fresh data

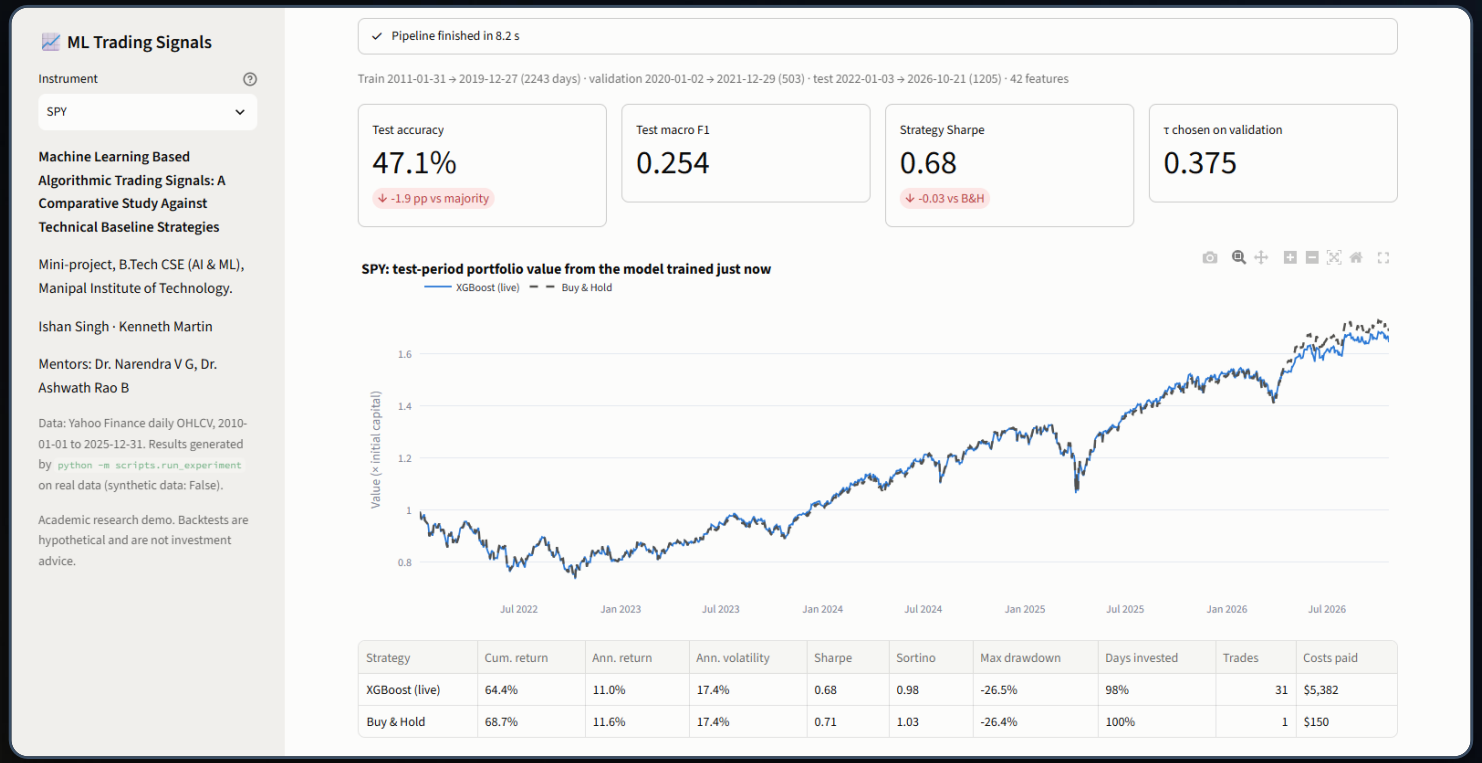


Fresh download → retrain → backtest in 8.2 s; opened on the next slide.

REAL SCREENSHOT · ONE RUN, NOT PART OF THE REPORTED RESULTS

Retrain live, with ten more months of data

TRAIN A MODEL LIVE SPY · XGBoost · the study's tuned settings



LIVE RUN · 24 OCT 2026

8.2 s

fresh download → features → split → XGBoost with the study's settings → τ → backtest (no re-tuning)

1,205

test days to 21 Oct 2026, 204 more than the study's test set

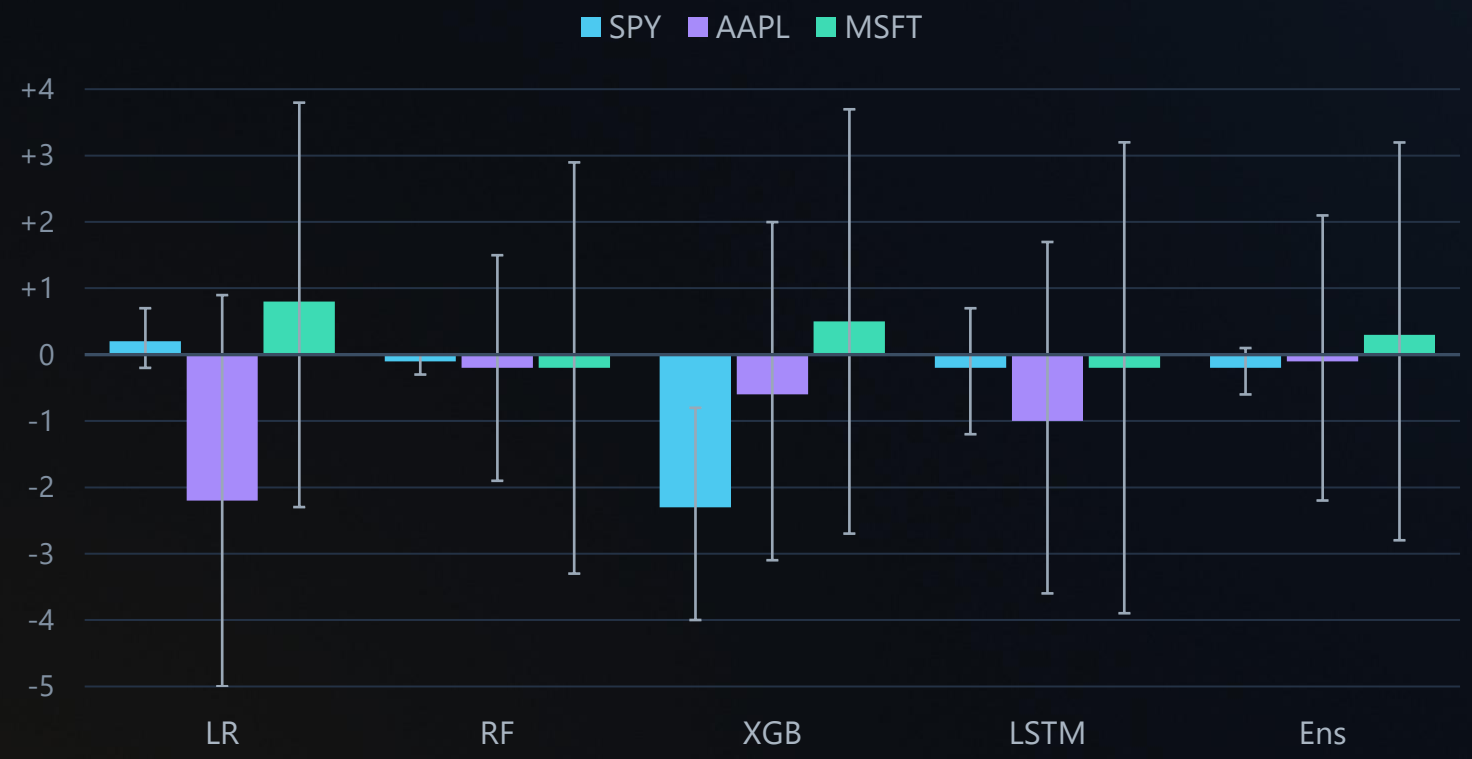
0.68 vs 0.71

Sharpe over the whole live window: XGBoost vs buy-and-hold

Consistent with the results that follow.

Accuracy: none reliably beat 'always predict Up'

Test accuracy minus 'always predict Up', percentage points (95% bootstrap CI)



47.2–51.0%

test accuracy, 5 models × 3 tickers

49.5–50.2%

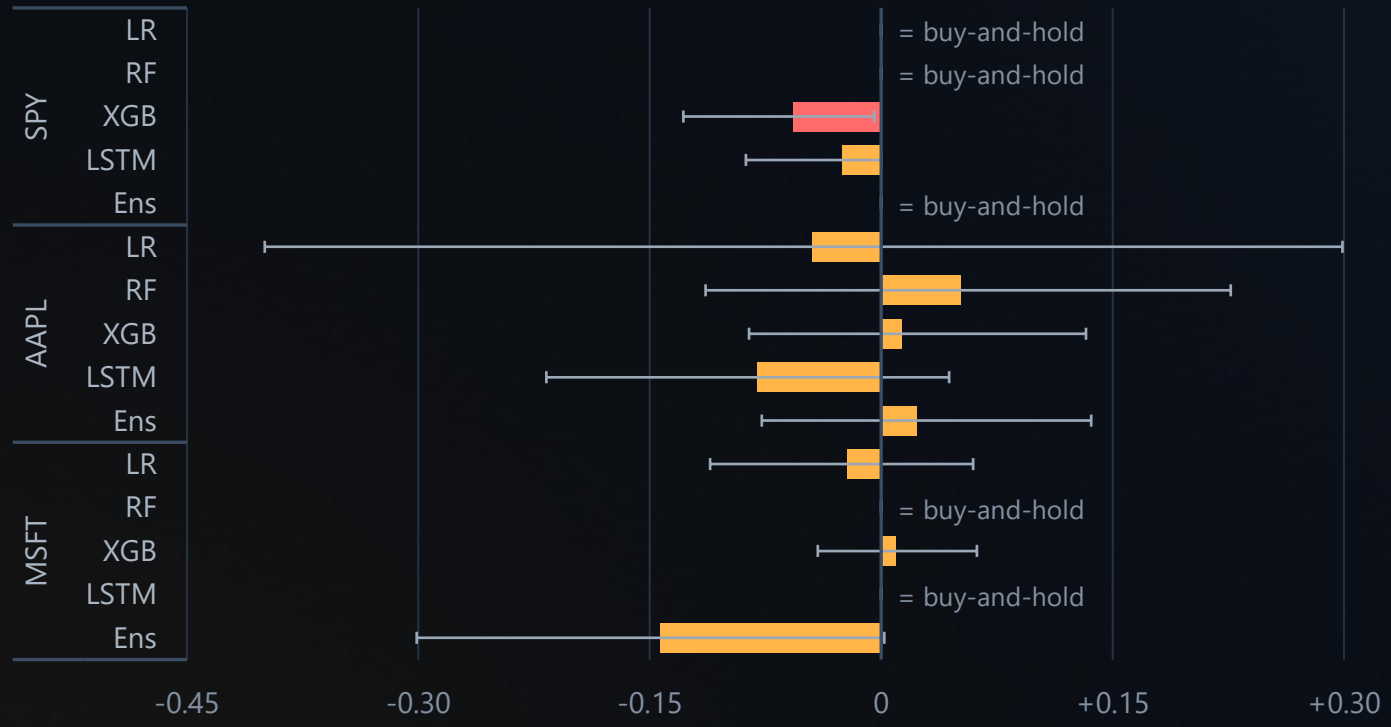
accuracy of 'always predict Up', the majority class

1 / 15

differs reliably from that baseline: SPY XGBoost, 2.3 pp worse

Trading: none of the 15 reliably beat buy-and-hold

τ-rule strategies: Sharpe minus buy-and-hold, test 2022–25 (95% bootstrap CI)



4 / 15

above buy-and-hold on Sharpe; all of their 95% intervals include zero

5 / 15

identical to buy-and-hold: the validation-chosen τ kept them invested on every test day

−48%

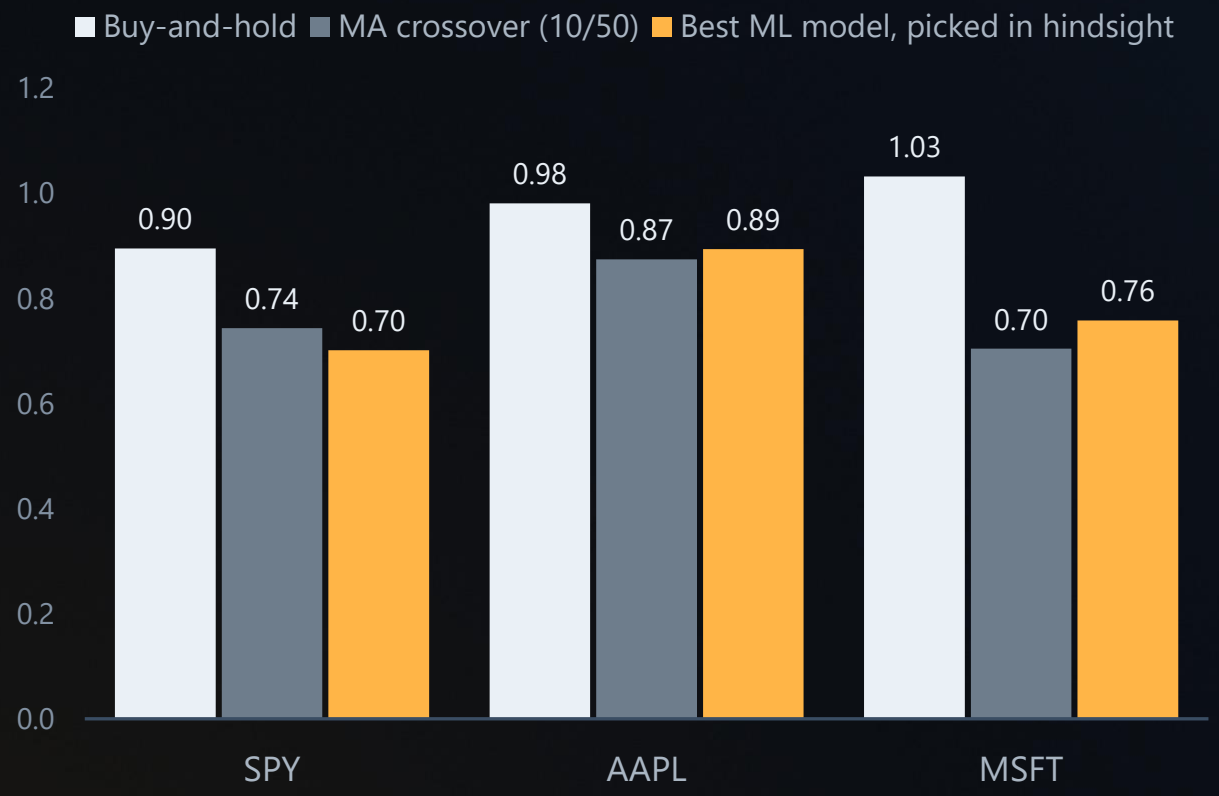
naive persistence on SPY, after 492 trades and costs

■ coral: 95% interval excludes zero (reliably worse)

Argmax rule (slide 13): 2 of 15 above buy-and-hold (SPY LR 0.76 vs 0.66, MSFT LR 0.60 vs 0.51); no intervals computed.

Walk-forward and robustness checks agree

Walk-forward 2016–2025: Sharpe ratio, retrained every year, out-of-sample



0 / 15

ML strategies beat buy-and-hold over the ten walk-forward years

ROBUSTNESS CHECKS · HOLD-OUT TEST 2022–2025

COSTS 7 / 15 beat buy-and-hold at 0 bps, 0 / 15 at 40 bps

SIZING 15% volatility targeting cut max drawdown by 21–44% at a similar Sharpe, by holding less

ASSUMPTIONS Binary target or close fills (no LSTM, reduced grids): 3–5 of 12 beat buy-and-hold, and the winners change

Challenges we handled, limitations we report

CHALLENGES WE HANDLED

01 **Leakage at every step**
Purged splits, point-in-time features and tests that fail on any look-ahead.

02 **Thresholds collapsing to buy-and-hold**
A bullish 2020–21 validation often picks 'always invested'; we report it rather than hide it.

03 **An unstable LSTM on ~2,200 training days**
A small one-layer network, early stopping on a purged slice and fixed seeds.

04 **Unfinished live bars**
Live data can include today's partial session; a guard drops it before 16:15 New York time.

05 **A 24-minute experiment**
Reduced grids for walk-forward, and a separate results folder for every flag combination.

LIMITATIONS WE REPORT

01 **Weak daily signal**
Liquid large caps have near-zero daily autocorrelation; ~50% accuracy is expected.

02 **Survivorship bias**
SPY, AAPL and MSFT were chosen in hindsight; weaker or delisted stocks could differ.

03 **Multiple testing**
Many models × thresholds × assets, and no deflated-Sharpe correction.

04 **Simple cost model**
A flat 15 bps; no market impact, spreads, borrow costs or taxes.

05 **Price and volume only**
No news, macro, fundamental or order-book data.

Where this goes next

THE STRETCH GOAL, DESIGNED LEAKAGE-SAFE · NEWS SENTIMENT VIA RETRIEVAL (RAG)

01
Time-stamped news
GDELT, SEC 8-K filings, licensed archives

→

02
Retrieve before close t
vector index by ticker and publication time

→

03
Score sentiment
a financial model such as FinBERT

→

04
Daily features
mean, count, spread; lagged one bar

→

05
Same protocol
same splits, τ rule and backtester

DATA

A broader, fairer universe

Include delisted stocks to remove survivorship bias, and test weekly and intraday horizons.

STATISTICS

Stronger evidence

Deflated Sharpe ratio and multiple-testing control; calibrate probabilities before thresholding.

PRACTICE

From backtest to live

Host the dashboard publicly and paper-trade the signals on live data with risk-based sizing.

What we can claim

1

ML did not reliably beat simple baselines at a daily horizon.

No model reliably beat 'always Up' on accuracy (one was reliably worse); τ-rule: 0 of 15 beat buy-and-hold in walk-forward.

2

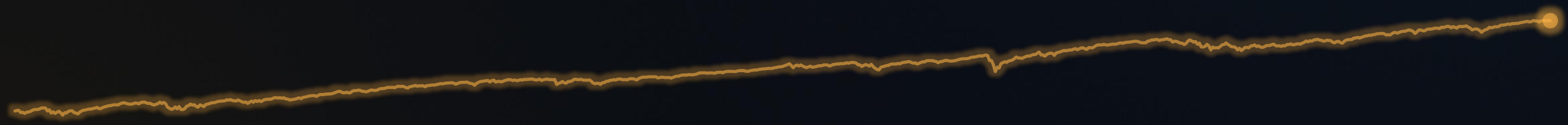
The contribution is the framework, not a trading edge.

Leakage-safe, cost-aware, covered by 301 tests and reproducible with one command.

3

Position sizing changed risk more than any forecast changed returns.

Volatility targeting cut max drawdown by 21–44% at a similar Sharpe ratio, by holding less, with no forecast at all.



Thank you

Questions and discussion

CODE `github.com/KennethMartin06/ml-based-algorithmic-trading-signals`

DEMO `streamlit run app/streamlit_app.py`

TEAM Ishan Singh · Kenneth Martin · MIT Manipal

